

```
import pandas as pd
```

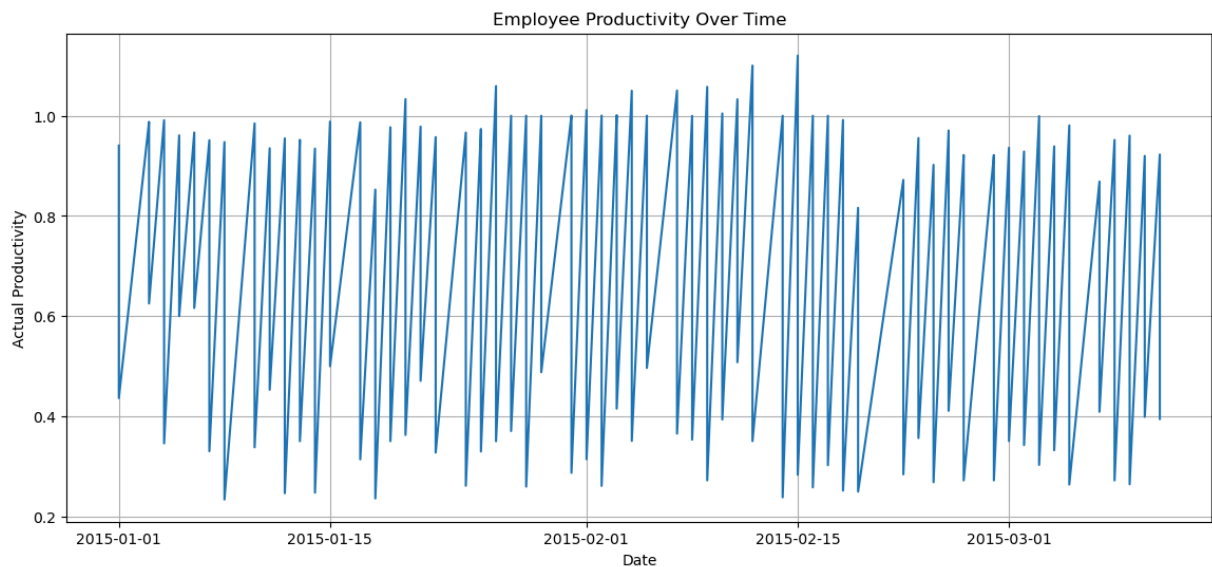
```
df = pd.read_csv("productivity+prediction+of+garment+employees.zip")
```

```
print(df.head())
```

```
In [3]: df['date'] = pd.to_datetime(df['date'])  
  
print(df.dtypes)
```

date	datetime64[ns]
quarter	object
department	object
day	object
team	int64
targeted_productivity	float64
smv	float64
wip	float64
over_time	int64
incentive	int64
idle_time	float64
idle_men	int64
no_of_style_change	int64
no_of_workers	float64
actual_productivity	float64
dtype:	object

```
In [4]: import matplotlib.pyplot as plt  
  
plt.figure(figsize=(14,6))  
  
plt.plot(  
    df['date'],  
    df['actual_productivity']  
)  
  
plt.title('Employee Productivity Over Time')  
plt.xlabel('Date')  
plt.ylabel('Actual Productivity')  
plt.grid(True)  
  
plt.show()
```



```
In [5]: mean_val = df['actual_productivity'].mean()
median_val = df['actual_productivity'].median()
std_val = df['actual_productivity'].std()

print("Mean:", mean_val)
print("Median:", median_val)
print("Standard Deviation:", std_val)
```

Mean: 0.7350910969791145

Median: 0.773333333

Standard Deviation: 0.17448790350939525

```
In [6]: weekly_avg = (
    df.set_index('date')
      .resample('W')['actual_productivity']
      .mean()
)

print(weekly_avg.head())
```

date

2015-01-04 0.768810

2015-01-11 0.774310

2015-01-18 0.731440

2015-01-25 0.723150

2015-02-01 0.788243

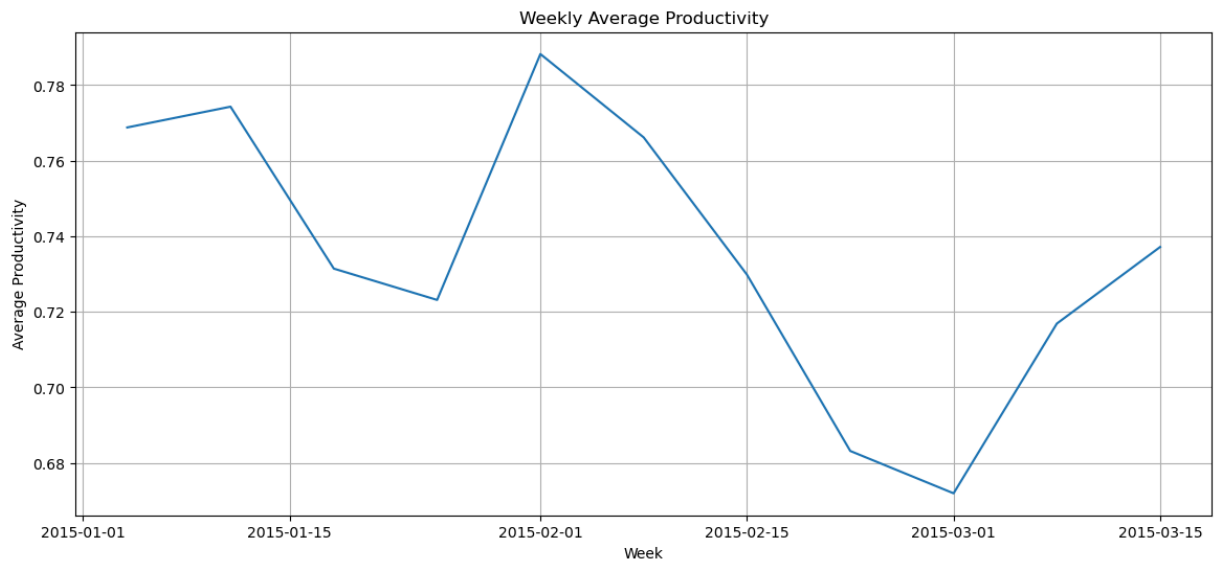
Freq: W-SUN, Name: actual_productivity, dtype: float64

```
In [7]: plt.figure(figsize=(14,6))

plt.plot(
    weekly_avg.index,
    weekly_avg.values
)

plt.title('Weekly Average Productivity')
plt.xlabel('Week')
plt.ylabel('Average Productivity')
plt.grid(True)
```

```
plt.show()
```



```
In [8]: from pymongo import MongoClient

client = MongoClient("mongodb://localhost:27017/")

db = client["garment_productivity_db"]

weekly_collection = db["weekly_productivity"]

weekly_df = weekly_avg.reset_index()
weekly_df.columns = ['Week', 'Average_Productivity']

records = weekly_df.to_dict("records")

weekly_collection.delete_many({})
weekly_collection.insert_many(records)

print("Weekly averages stored successfully.")
```

Weekly averages stored successfully.

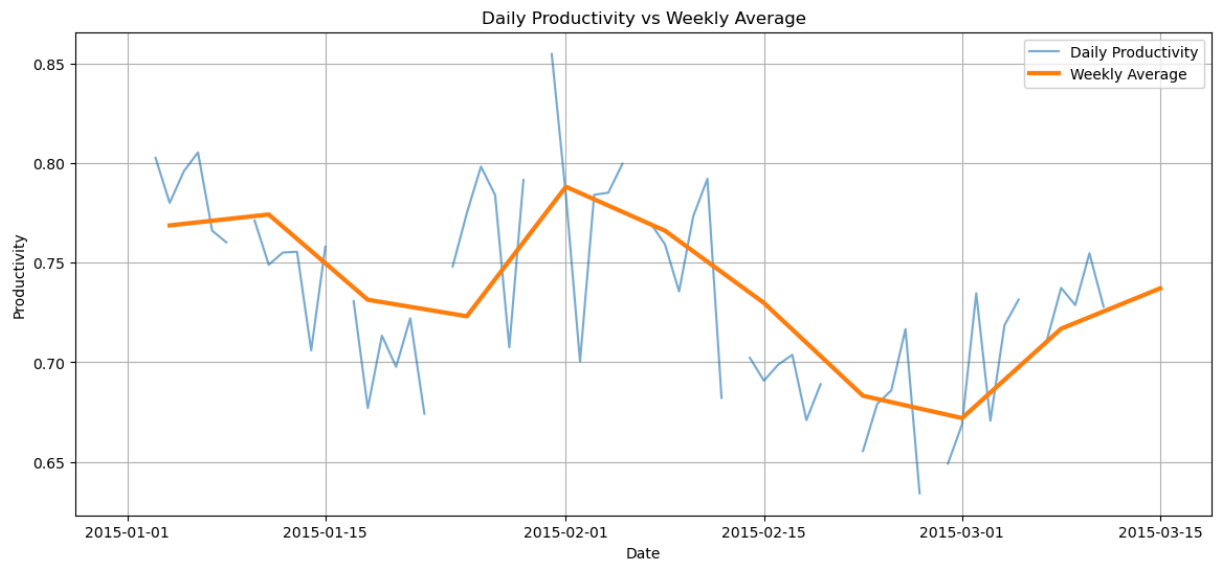
```
In [9]: daily_avg = (
    df.set_index('date')
      .resample('D')['actual_productivity']
      .mean()
)

plt.figure(figsize=(14,6))

plt.plot(
    daily_avg.index,
    daily_avg.values,
    label='Daily Productivity',
    alpha=0.6
)

plt.plot(
```

```
weekly_avg.index,  
weekly_avg.values,  
linewidth=3,  
label='Weekly Average'  
)  
  
plt.title('Daily Productivity vs Weekly Average')  
plt.xlabel('Date')  
plt.ylabel('Productivity')  
plt.legend()  
plt.grid(True)  
  
plt.show()
```



The productivity values fluctuate over time. Weekly averaging smooths short-term variations and helps reveal the underlying productivity trend. The weekly average line provides a clearer representation of employee productivity patterns compared to the original daily observations.

In []: